

Non-Accidental Trauma



Section I: Scenario Demographics

Scenario Title:	Non Accidental Injury- Shaken Baby		
Date of Development:	(11/01/2018)		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Suzan Schneeweiss
Affiliations/Institution(s):	Sick Kids Learning Institute
Contact E-mail (optional):	

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To review the management of a seizing infant with non-accidental head injury
CRM Objectives:	1. Effectively lead a team through care of a critically ill infant 2. Demonstrate high quality closed-loop communication during resuscitation 3. Sensitively address the patient's mother and notify re: team concerns regarding non-accidental injury
Medical Objectives:	1. Rapidly estimate pediatric weight for medication dosing and resuscitation equipment 2. Manage seizures in the newborn 3. Recognize the presentation of possible non-accidental injury 4. Lead team through intubation after airway maneuvers unsuccessful

Case Summary: Brief Summary of Case Progression and Major Events
The team has been called to help in the ED after a 1 month-old male is brought in seizing. The team is expected to manage the seizure, but then will subsequently realize on examination there are concerning signs for non-accidental trauma, specifically head injury. The team will be expected to establish definitive airway management and consult with PICU and local child protection services.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.
https://www.cps.ca/en/documents/position/convulsive-status-epilepticus

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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates		Brief Description of Role	
Nurse		To assist at bedside, describe infant's status, and provide history.	
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Neonatal Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ Neonatal ET Tubes	☒ Other: Radiant warmer	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Boggy swelling left parietal area			
Bruising on back			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	15 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 1 month-old male is brought into the ED due to poor feeding and lethargy. The baby was apparently well until this morning, when his mom noticed it was difficult to wake and feed him. There has been no fever. The baby vomited once this morning, and is voiding and stooling normally.

The nurse in triage notices abnormal movements and brings the baby in to your team in the resuscitation room.

B. Patient Profile and History

Patient Name: Cody	Age: 30 days	Weight: 4 kg
Gender: ☒ M ☐ F		
Chief Complaint: Seizing baby		
History of Presenting Illness: Poor feeding and lethargy x 1 day, starts seizing at triage		
Past Medical History:	Full term 39 weeks GBS negative No PROM	Medications: None

Allergies: None

Social History: Lives with mother and her boyfriend. Mother attends college part time.

Family History: None

Review of Systems:	CNS:	Difficult to wake baby for feeds today		
	HEENT:	Nil		
	CVS:	Nil		
	RESP:	Nil		
	GI:	Vomited once this morning. Stooling normally.		
	GU:	Normal urine output.		
	MSK:	Nil	INT:	Nil.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 180/min	BP: 80/p	RR: 22/min O ₂ SAT: 92%
Rhythm: Sinus with background artifact	T: 36.5°C	Glucose: 2.0 mmol/L
General Status: Actively seizing		
CNS:	Pupils dilated, sluggish, tonic clonic movement of 4 limbs	
HEENT:	Boggy swelling over left parietal area	
CVS:	Sinus tachycardia, HR 160	
RESP:	Shallow respirations	
ABDO:	Soft, NT	
GU:	Nil	
MSK:	Nil	SKIN: Bruises to back/buttocks area

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus tachycardia HR: 180/min BP: 80/p RR: 22/min O ₂ SAT: 92% T: 36.5°C	Actively seizing	<u>Learner Actions</u> - ☐ Apply 100% O ₂ , monitors - ☐ Obtain IV access - ☐ Labs including tox screen and blood cultures - ☐ Check cap sugar: 2.0 - ☐ Obtains weight from RN (4 kg), or uses Broselow tape - ☐ IV D10W, 5ml/kg (20ml) - ☐ IV Ativan 0.1 mg/kg, or IM midazolam 0.2 mg/kg - ☐ Asks for overbed warmer/radiant warmer - ☐ Call RT	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Only D10 given → seizure continues - If no overbed warmer, RN to cue "he's starting to feel cold" <u>Triggers</u> <i>For progression to next state</i> - Benzos + D10 given → 2. Post ictal
2. Post-ictal HR: 160/min BP: 80/p RR: 6, shallow O ₂ SAT: 80%	Pale, lethargic, grunting. No longer seizing.	<u>Learner Actions</u> - ☐ Jaw thrust, suction airway - ☐ Call RT (if not yet done) - ☐ Begin BVM at rate 20 - ☐ IV NS bolus 10-20 cc/kg - ☐ Re-Check cap sugar: 4.4 - ☐ IV Ampicillin (50 mg/kg), Cefotaxime (50 mg/kg), and acyclovir (20mg/kg) - ☐ Recognize signs of head injury/non-accidental trauma - ☐ IV phenobarbital (20mg/kg) - ☐ Prepare for intubation	<u>Modifiers</u> - No jaw thrust, O ₂ → SpO ₂ 70% - Jaw thrust, O ₂ → SpO ₂ 85% - RN to prompt: "What's with this bruising on his head? I noticed something on his back, too." <u>Triggers</u> - Preparing to intubate → 3. Persistent apnea - 5 minutes into state → 3. Persistent apnea
3. Persistent apnea HR: 140/min BP: 80/p RR: 0 O ₂ SAT: 80%		<u>Learner Actions</u> - ☐ Prepare for intubation - ☐ Ketamine 1.5-2mg/kg OR - ☐ Fentanyl 1.5-2 mcg/kg - ☐ Rocuronium 0.6-1mg/kg - ☐ Atropine (0.02 mg/kg) at bedside or given, Epi at bedside - ☐ Appropriate size tube (3 cuffed, 3.5 uncuffed) and tube depth (9 cm)	<u>Modifiers</u> - No plan to intubate → RN to trigger "how long are we going to continue bagging for?" <u>Triggers</u> - Successful intubation → 4. Disposition
4. Disposition HR: 140 BP: 80/p RR: 20 (vented) O ₂ SAT: 95%		<u>Learner Actions</u> - ☐ Post intubation CXR - ☐ Initiate sedation - ☐ Arrange CT head - ☐ Discuss trauma concerns with parent - ☐ Consult PICU - ☐ Call child protective services	END CASE PRN



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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No bloodwork is provided during the case.

Images (ECGs, CXRs, etc.)



(CXR source: <https://radiopaedia.org/articles/neonatal-pneumonia>)

Ultrasound Video Files (if applicable)

None required.

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1. What are some common non-accidental injury patterns in pediatrics? 2. What is in the differential diagnosis for a seizing infant? 3. If IV access was not successful, what other routes and doses of anti-epileptic medications could you use? 4. How did the decision to intubate feel? Were all members on the same page? Did the intubation run smoothly? 5. How did it feel to raise concerns for non-accidental trauma with the parent?	
Key Moments	
Recognition of seizing infant and need check glucose and start first line anti-epileptics	
Recognition of apnea and establishment of definitive airway	
Recognition of non accidental injury and appropriate consultation	